

ECE 4020-5020: Digital Signal Processing

Homework 3 (100 pts.)

Due on Thursday, March 26, 2026

Problem 1 (25 pts.):

Determine the Fourier spectral coefficients c_k and use them to determine the signal power for each of the following discrete-time periodic signals,

a. $x(n) = \{\cdots, -1, 2, \underset{\uparrow}{1}, 2, -1, 0, -1, 2, 1, 2, \cdots\}$

b. $x(n) = 4 \sin\left(\frac{\pi(n-2)}{3}\right)$

c. $x(n) = \cos\left(\frac{2\pi}{3}n\right) + \sin\left(\frac{2\pi}{5}n\right)$

Hint Part 1(c): The period of the sum of periodic sinusoids is the least common multiple of the individual periods.

Problem 2 (20 pts.):

Determine the discrete-time periodic signal $x(n)$ from its Fourier coefficients.

a. $c_k = \{\cdots, 0, \frac{1}{4}, \frac{1}{2}, 1, \underset{\uparrow}{2}, 1, \frac{1}{2}, \frac{1}{4}, 0, \cdots\}$ for period $N=8$

b. $c_k = \{\frac{9}{6}, \frac{4}{6}, 0, \frac{1}{6}, 0, \frac{4}{6}\}$ for period $N=6$

c. $c_k = \begin{cases} \sin \frac{k\pi}{3}, & 0 \leq k \leq 6 \\ 0, & k = 7 \end{cases}$ for period $N=8$

Problem 3 (15 pts.):

Determine the Fourier transform of the following discrete-time aperiodic signals.

a. $x(n) = \left(\frac{1}{4}\right)^n u(n+4)$

b. $x(n) = \begin{cases} 2 - \left(\frac{1}{2}\right)^n, & |n| \leq 4 \\ 0, & \text{otherwise} \end{cases}$

c. $x(n) = \{-2, -1, \underset{\uparrow}{0}, 1, 2\}$

Problem 4 (15 pts.):

For a discrete-time signal $x(n]$ with Fourier transform $X(\omega) = \frac{1}{1 - ae^{-j\omega}}$, use Fourier transform properties to determine the Fourier transforms of the following signals,

a. $x(2n+1)$

b. $x(n) \cos(0.3\pi n)$

c. $x(n) * x(-n)$

Problem 5 (25 pts.):

Determine the following for an LTI system whose impulse response is $h(n) = \left(\frac{1}{2}\right)^n u(n)$.

a. Expressions for the frequency response $H(\omega)$, magnitude response $|H(\omega)|$, and phase response $\angle H(\omega)$.

b. Expression for the output response $y(n]$ when the input is $x(n) = \cos\left(\frac{3\pi}{10}n\right)$.

ECE 5020 Homework 3

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Problem 1

For a period- N sequence,

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j \frac{2\pi}{N} kn}, \quad x(n) = \sum_{k=0}^{N-1} c_k e^{j \frac{2\pi}{N} kn},$$

and Parseval gives

$$P = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2 = \sum_{k=0}^{N-1} |c_k|^2.$$

(a) $x(n) = \{\dots, -1, 2, 1 \uparrow, 2, -1, 0, \dots\}$

One period is $\{1, 2, -1, 0, -1, 2\}$, so $N = 6$. By direct evaluation, it follows that:

$c_0 = \frac{1}{2},$	$c_1 = \frac{2}{3},$	$c_2 = 0,$	$c_3 = -\frac{5}{6},$	$c_4 = 0,$	$c_5 = \frac{2}{3}.$
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Hence, it follows that

$P = \sum_{k=0}^5 c_k ^2 = \frac{1}{4} + \frac{4}{9} + 0 + \frac{25}{36} + 0 + \frac{4}{9} = \frac{11}{6}.$
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Now, we need to check in the time domain:

$$\frac{1}{6} (1^2 + 2^2 + (-1)^2 + 0^2 + (-1)^2 + 2^2) = \frac{11}{6}.$$

(b) $x(n) = 4 \sin\left(\frac{\pi(n-2)}{3}\right)$

The fundamental frequency is

$$\omega_0 = \frac{\pi}{3} = \frac{2\pi}{6},$$

so $N = 6$. By using Euler's identity,

$$x(n) = \frac{4}{2j} \left(e^{j \frac{\pi}{3}(n-2)} - e^{-j \frac{\pi}{3}(n-2)} \right) = \frac{2}{j} \left(e^{j \frac{2\pi}{6} n} e^{-j \frac{2\pi}{3}} - e^{-j \frac{2\pi}{6} n} e^{j \frac{2\pi}{3}} \right).$$

Then the only nonzero DTFS coefficients are

$c_1 = \frac{2}{j} e^{-j \frac{2\pi}{3}} = -\sqrt{3} + j, \quad c_5 = c_{-1} = c_1^* = -\sqrt{3} - j.$ $P = c_1 ^2 + c_5 ^2 = 4 + 4 = 8.$

(c) $x(n) = \cos\left(\frac{2\pi}{3}n\right) + \sin\left(\frac{2\pi}{5}n\right)$

The periods of the two sinusoids are 3 and 5, so

$$N = \text{lcm}(3, 5) = 15, \quad \omega_0 = \frac{2\pi}{15}.$$

Since

$$\frac{2\pi}{3} = 5\omega_0, \quad \frac{2\pi}{5} = 3\omega_0,$$

we get

$$\begin{aligned} \cos\left(\frac{2\pi}{3}n\right) &\Rightarrow c_5 = \frac{1}{2}, \quad c_{10} = \frac{1}{2}, \\ \sin\left(\frac{2\pi}{5}n\right) &\Rightarrow c_3 = \frac{1}{2j} = -\frac{j}{2}, \quad c_{12} = -\frac{1}{2j} = \frac{j}{2}. \end{aligned}$$

Thus

$$\boxed{\begin{aligned} c_3 &= -\frac{j}{2}, \quad c_5 = \frac{1}{2}, \quad c_{10} = \frac{1}{2}, \quad c_{12} = \frac{j}{2}, \\ P &= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} + \frac{1}{4} = 1. \end{aligned}}$$

with all other $c_k = 0$.

Problem 2

If we use the DTFS synthesis formula, it follows that:

$$x(n) = \sum_{k=0}^{N-1} c_k e^{j\frac{2\pi}{N}kn}.$$

(a) $c_k = \{\dots, 0, \frac{1}{4}, \frac{1}{2}, 1 \uparrow, 2, 1, \frac{1}{2}, \frac{1}{4}, 0, \dots\}$, $N = 8$

From the arrow location,

$$\boxed{c_0 = 1, \quad c_1 = 2, \quad c_2 = 1, \quad c_3 = \frac{1}{2}, \quad c_4 = \frac{1}{4}, \quad c_5 = 0, \quad c_6 = \frac{1}{4}, \quad c_7 = \frac{1}{2}.$$

These coefficients are real, but they are not conjugate-symmetric because, for example,

$$c_1 = 2 \neq c_7 = \frac{1}{2}.$$

So the signal is generally complex-valued, and we should use the synthesis sum directly:

$$\boxed{x(n) = 1 + 2e^{j\pi n/4} + e^{j\pi n/2} + \frac{1}{2}e^{j3\pi n/4} + \frac{1}{4}e^{j\pi n} + \frac{1}{4}e^{j3\pi n/2} + \frac{1}{2}e^{j7\pi n/4}.$$

An equivalent cosine/sine form is

$$\boxed{\begin{aligned} x(n) &= 1 + \frac{5}{2} \cos\left(\frac{\pi n}{4}\right) + j\frac{3}{2} \sin\left(\frac{\pi n}{4}\right) + \frac{5}{4} \cos\left(\frac{\pi n}{2}\right) + j\frac{3}{4} \sin\left(\frac{\pi n}{2}\right) \\ &\quad + \frac{1}{2} \cos\left(\frac{3\pi n}{4}\right) + j\frac{1}{2} \sin\left(\frac{3\pi n}{4}\right) + \frac{1}{4}(-1)^n. \end{aligned}}$$

(b) $c_k = \left\{\frac{9}{6}, \frac{4}{6}, 0, \frac{1}{6}, 0, \frac{4}{6}\right\}$, $N = 6$

Here

$$c_0 = \frac{3}{2}, \quad c_1 = \frac{2}{3}, \quad c_2 = 0, \quad c_3 = \frac{1}{6}, \quad c_4 = 0, \quad c_5 = \frac{2}{3}.$$

Because $c_5 = c_1$ and the coefficients are real,

$$x(n) = c_0 + 2c_1 \cos\left(\frac{2\pi}{6}n\right) + c_3 e^{j\pi n}.$$

Since $e^{j\pi n} = (-1)^n$,

$$x(n) = \frac{3}{2} + \frac{4}{3} \cos\left(\frac{\pi n}{3}\right) + \frac{1}{6}(-1)^n.$$

(c) $c_k = \sin\left(\frac{k\pi}{3}\right)$ for $0 \leq k \leq 6$, $c_7 = 0$, $N = 8$

Evaluating the coefficients,

$$c_0 = 0, \quad c_1 = \frac{\sqrt{3}}{2}, \quad c_2 = \frac{\sqrt{3}}{2}, \quad c_3 = 0, \quad c_4 = -\frac{\sqrt{3}}{2}, \quad c_5 = -\frac{\sqrt{3}}{2}, \quad c_6 = 0, \quad c_7 = 0.$$

Since the coefficients are not conjugate-symmetric, the signal is generally complex-valued. Direct substitution gives

$$x(n) = \frac{\sqrt{3}}{2} \left(e^{j\pi n/4} + e^{j\pi n/2} - e^{j\pi n} - e^{j5\pi n/4} \right), \quad n = 0, 1, \dots, 7.$$

Problem 3

The DTFT is

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}.$$

(a) $x(n) = \left(\frac{1}{4}\right)^n u(n+4)$

Since $u(n+4) = 1$ for $n \geq -4$,

$$X(\omega) = \sum_{n=-4}^{\infty} \left(\frac{1}{4}\right)^n e^{-j\omega n}.$$

This is a geometric series with ratio $\frac{1}{4}e^{-j\omega}$, so

$$\begin{aligned} X(\omega) &= \frac{\left(\frac{1}{4}\right)^{-4} e^{j4\omega}}{1 - \frac{1}{4}e^{-j\omega}} = \frac{256e^{j4\omega}}{1 - \frac{1}{4}e^{-j\omega}}. \\ X(\omega) &= \frac{1}{1 - \frac{1}{4}e^{-j\omega}} + 4e^{j\omega} + 16e^{j2\omega} + 64e^{j3\omega} + 256e^{j4\omega}. \end{aligned}$$

Equivalently,

(b) $x(n) = 2 - \left(\frac{1}{2}\right)^n$ for $|n| \leq 4$, and 0 otherwise

Split the transform into two finite sums:

$$X(\omega) = 2 \sum_{n=-4}^4 e^{-j\omega n} - \sum_{n=-4}^4 \left(\frac{1}{2}\right)^n e^{-j\omega n}.$$

For the first sum,

$$\sum_{n=-4}^4 e^{-j\omega n} = \frac{\sin(9\omega/2)}{\sin(\omega/2)}.$$

For the second sum, let

$$r = \frac{1}{2}e^{-j\omega}.$$

Then

$$\sum_{n=-4}^4 r^n = r^{-4} \frac{1 - r^9}{1 - r} = 16e^{j4\omega} \frac{1 - \frac{1}{512}e^{-j9\omega}}{1 - \frac{1}{2}e^{-j\omega}}.$$

Therefore

$$X(\omega) = 2 \frac{\sin(9\omega/2)}{\sin(\omega/2)} - 16e^{j4\omega} \frac{1 - \frac{1}{512}e^{-j9\omega}}{1 - \frac{1}{2}e^{-j\omega}}.$$

(c) $x(n) = \{-2, -1, 0 \uparrow, 1, 2\}$

Here $x(n) = n$ for $-2 \leq n \leq 2$, so

$$X(\omega) = \sum_{n=-2}^2 ne^{-j\omega n} = -2e^{j2\omega} - e^{j\omega} + e^{-j\omega} + 2e^{-j2\omega}.$$

Thus

$$X(\omega) = (e^{-j\omega} - e^{j\omega}) + 2(e^{-j2\omega} - e^{j2\omega}) = -2j \sin \omega - 4j \sin(2\omega).$$

Hence

$$X(\omega) = -2j(\sin \omega + 2 \sin 2\omega).$$

Problem 4

Given

$$X(\omega) = \frac{1}{1 - ae^{-j\omega}},$$

use DTFT properties.

(a) $x(2n + 1)$

Let $y(n) = x(n + 1)$. Then by time shifting,

$$Y(\omega) = e^{j\omega} X(\omega) = \frac{e^{j\omega}}{1 - ae^{-j\omega}}.$$

Now $x(2n + 1) = y(2n)$. By the downsampling property,

$$\mathcal{F}\{y(2n)\} = \frac{1}{2} \left[Y\left(\frac{\omega}{2}\right) + Y\left(\frac{\omega}{2} - \pi\right) \right].$$

So

$$\mathcal{F}\{x(2n+1)\} = \frac{e^{j\omega/2}}{2} \left[\frac{1}{1 - ae^{-j\omega/2}} - \frac{1}{1 + ae^{-j\omega/2}} \right] = \frac{e^{j\omega/2}}{2} \cdot \frac{2ae^{-j\omega/2}}{1 - a^2e^{-j\omega}}.$$

$$\boxed{\mathcal{F}\{x(2n+1)\} = \frac{a}{1 - a^2e^{-j\omega}}.}$$

(b) $x(n) \cos(0.3\pi n)$

Using the modulation property,

$$\mathcal{F}\{x(n) \cos(0.3\pi n)\} = \frac{1}{2}X(\omega - 0.3\pi) + \frac{1}{2}X(\omega + 0.3\pi).$$

Hence

$$\boxed{\mathcal{F}\{x(n) \cos(0.3\pi n)\} = \frac{1}{2} \cdot \frac{1}{1 - ae^{-j(\omega-0.3\pi)}} + \frac{1}{2} \cdot \frac{1}{1 - ae^{-j(\omega+0.3\pi)}}.}$$

(c) $x(n) * x(-n)$

Time reversal gives

$$\mathcal{F}\{x(-n)\} = X(-\omega) = \frac{1}{1 - ae^{j\omega}}.$$

Convolution in time corresponds to multiplication in frequency, so

$$\mathcal{F}\{x(n) * x(-n)\} = X(\omega)X(-\omega) = \frac{1}{(1 - ae^{-j\omega})(1 - ae^{j\omega})}.$$

Using

$$(1 - ae^{-j\omega})(1 - ae^{j\omega}) = 1 - 2a \cos \omega + a^2,$$

we obtain

$$\boxed{\mathcal{F}\{x(n) * x(-n)\} = \frac{1}{1 - 2a \cos \omega + a^2}.}$$

Problem 5

The impulse response is

$$h(n) = \left(\frac{1}{2}\right)^n u(n).$$

(a) $H(\omega)$, $|H(\omega)|$, and $\angle H(\omega)$

The frequency response is the causal geometric DTFT:

$$H(\omega) = \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n e^{-j\omega n} = \frac{1}{1 - \frac{1}{2}e^{-j\omega}}.$$

For the magnitude,

$$\left|1 - \frac{1}{2}e^{-j\omega}\right|^2 = \left(1 - \frac{1}{2}\cos \omega\right)^2 + \left(\frac{1}{2}\sin \omega\right)^2 = \frac{5}{4} - \cos \omega.$$

Therefore

$$|H(\omega)| = \frac{1}{\sqrt{\frac{5}{4} - \cos \omega}}.$$

For the phase,

$$1 - \frac{1}{2}e^{-j\omega} = \left(1 - \frac{1}{2}\cos\omega\right) + j\frac{1}{2}\sin\omega,$$

so a branch-correct expression is

$$\boxed{\begin{aligned} H(\omega) &= \frac{1}{1 - \frac{1}{2}e^{-j\omega}}, & |H(\omega)| &= \frac{1}{\sqrt{\frac{5}{4} - \cos\omega}}, \\ \angle H(\omega) &= -\text{atan2}\left(\frac{1}{2}\sin\omega, 1 - \frac{1}{2}\cos\omega\right). \end{aligned}}$$

(b) Response to $x(n) = \cos\left(\frac{3\pi}{10}n\right)$

For an LTI system driven by a sinusoid,

$$y(n) = |H(\omega_0)| \cos(\omega_0 n + \angle H(\omega_0)), \quad \omega_0 = \frac{3\pi}{10}.$$

Thus

$$y(n) = \frac{1}{\sqrt{\frac{5}{4} - \cos(3\pi/10)}} \cos\left(\frac{3\pi}{10}n - \text{atan2}\left(\frac{1}{2}\sin(3\pi/10), 1 - \frac{1}{2}\cos(3\pi/10)\right)\right).$$

Numerically,

$$|H(\omega_0)| \approx 1.229, \quad \angle H(\omega_0) \approx -0.520 \text{ rad},$$

so

$$\boxed{y(n) \approx 1.229 \cos\left(\frac{3\pi}{10}n - 0.520\right)}.$$